

Baseline (0 hr): Chromatin modules, TF programs, and genomic context

Biclustering of the 0 hr signal revealed a strongly polarized center–flanks architecture (Figure 1). Across most modules (row clusters), the center column was selectively elevated while both flanks were reduced, consistent with a shared footprint of focal chromatin activity surrounded by comparatively quiescent edges. Several smaller modules deviated from this predominant center-high, flanks-low pattern, including left-skewed blocks with signal confined to the left column and rare inversions with depressed centers and relatively higher flanks, indicating additional, less frequent organizational modes at baseline. A minority of modules exhibited uniformly low signal across all columns.

Transcription factor (TF) motif enrichment segregated with these modules and nominated distinct candidate regulators (Figure 2). Enrichment was broadly higher in mid-to-late clusters, peaking across H–I and extending through J–N, and included CHD1/CHD2, RNA polymerase II (POLR2A), MAX, and multiple GATA factors, consistent with recruitment of class I CHD remodelers and the transcriptional machinery to accessible, transcriptionally engaged chromatin (Sun et al., 2026; Haque et al., 2026). By contrast, early clusters (A–C) showed weaker signals for several factors, with notable depletions for KAT2A at A and MYC at C, and reduced E2F4 and CTCF across A–D, suggesting a relative paucity of acetyltransferase activity and cell-cycle/insulator programs at baseline in these groups. GABPA showed broad enrichment with a pronounced peak in cluster A, aligning with its established role at constitutively accessible regulatory DNA and promoter-like ETS occupancy (Hollenhorst et al., 2009), although other promoter-associated factors were not uniformly elevated in the earliest groups. Together, these trends point to a progression from TF-poor, structurally accessible elements toward modules with stronger engagement of lineage (GATA), remodeling (CHD1/2), and transcriptional (POLR2A) programs.

ENCODE candidate cis-regulatory element (cCRE) annotations corroborated this regulatory continuum (Figure 3). Enhancer-like classes predominated overall, with distal enhancers (dELS) most represented in earlier groups (A–F) and a gradual increase in promoter-like content in later groups: promoter-like enhancers (pELS) rose from H–O and promoter-like signatures (PLS) expanded from G–O, especially J–O. DNase-only and TF-associated accessible classes (CA–TF) were more evident in A–C and E–F and became scarce thereafter, consistent with early clusters harboring unmodified or TF-marked accessibility and later clusters progressively accruing promoter-like chromatin features. Gene-based annotations further situated these modules (Figure 4): regions were predominantly intronic across clusters, with variable but smaller promoter and exon

contributions. Promoter overlap was relatively elevated in B, D, H, I, K, and O, but lower in A and J, while intergenic “other” was higher in A and D. These distributions indicate that the prevalent center-high modules are largely intronic/distal enhancer elements at baseline, with subsets transitioning toward promoter-enriched content in later clusters, mirroring the cCRE shift.

In aggregate, the 0 hr landscape comprises co-regulated chromatin modules defined by a center-high, flanks-low contrast and stratified along an enhancer-to-promoter axis. Motif enrichments highlight candidate regulators matched to this stratification: GATA factors, MAX-associated programs, CHD1/2 remodelers, and POLR2A in mid-to-late clusters that bear stronger promoter-like content and transcriptional engagement (Sun et al., 2026; Haque et al., 2026), and GABPA in early modules with pronounced ETS signatures (Hollenhorst et al., 2009). Depletion of KAT2A, E2F4, and CTCF in the earliest groups

further distinguishes these enhancer-dominated modules from later, promoter-increasing groups.

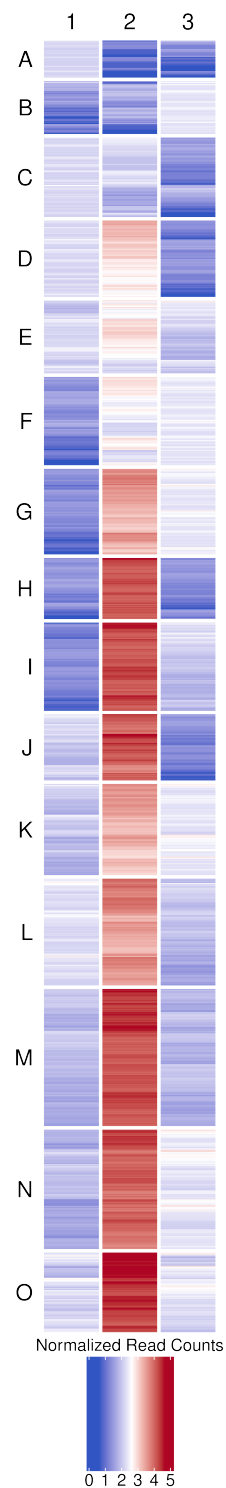


Figure 1. Biclustering heatmap reveals baseline chromatin signal modules. *Biclustering of the 0 hr count matrix organizes genomic regions (rows) and feature pairs (columns). Most modules show elevated activity in the middle column and reduced activity in the flanking columns, with a smaller number exhibiting uniformly low or inverse patterns.*

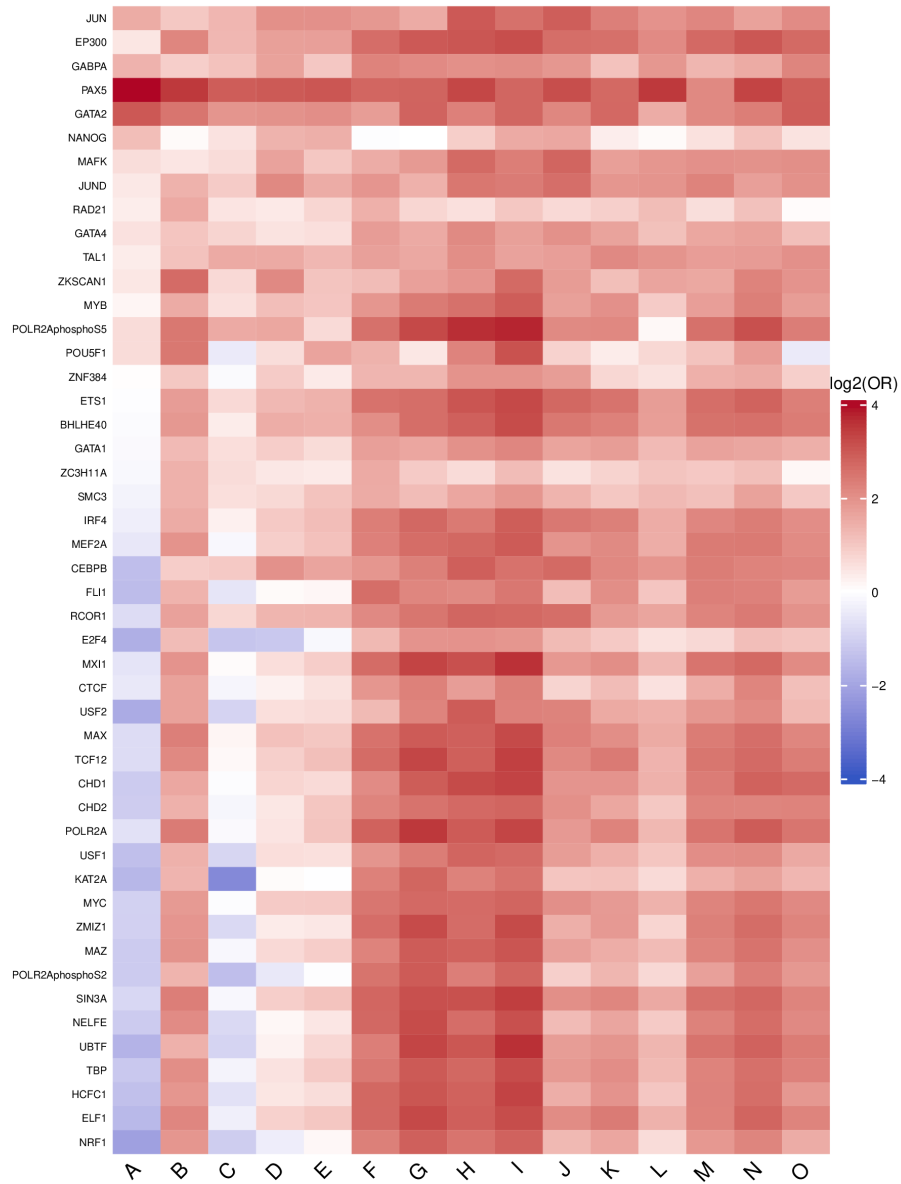


Figure 2. TF motif enrichment across 0hr region clusters. Motif overrepresentation ($\log_2$ odds ratio) is summarized for clusters A–O. Enrichment concentrates in central-right clusters, particularly around H–I, while several motifs show weaker or depleted signals in early clusters A–D. Broad enrichment is evident for factors such as GABPA, with localized depletion for KAT2A and MYC in early clusters.

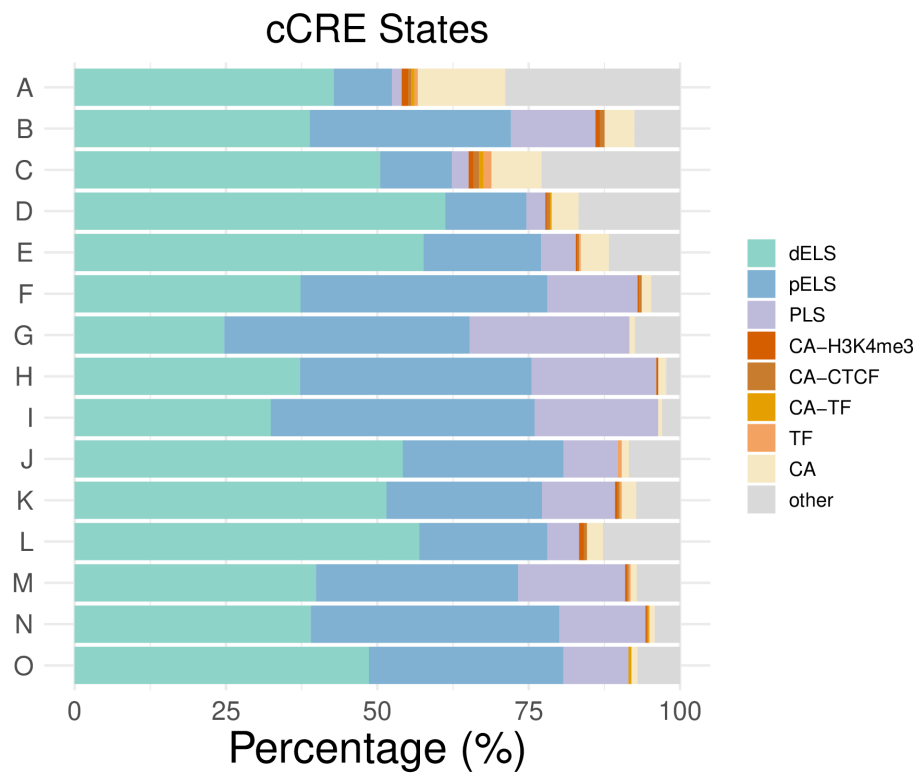


Figure 3. cCRE class composition across 0hr bicluster groups. Shown are cCRE annotations for groups A–O in the 0hr sample. Distal and proximal enhancer states constitute most regions overall, while promoter-like states are relatively higher in groups G–O; CA/TF-associated categories and other annotations contribute smaller shares. The pattern indicates a transition from distal enhancer enrichment in early groups toward greater promoter-like representation later.

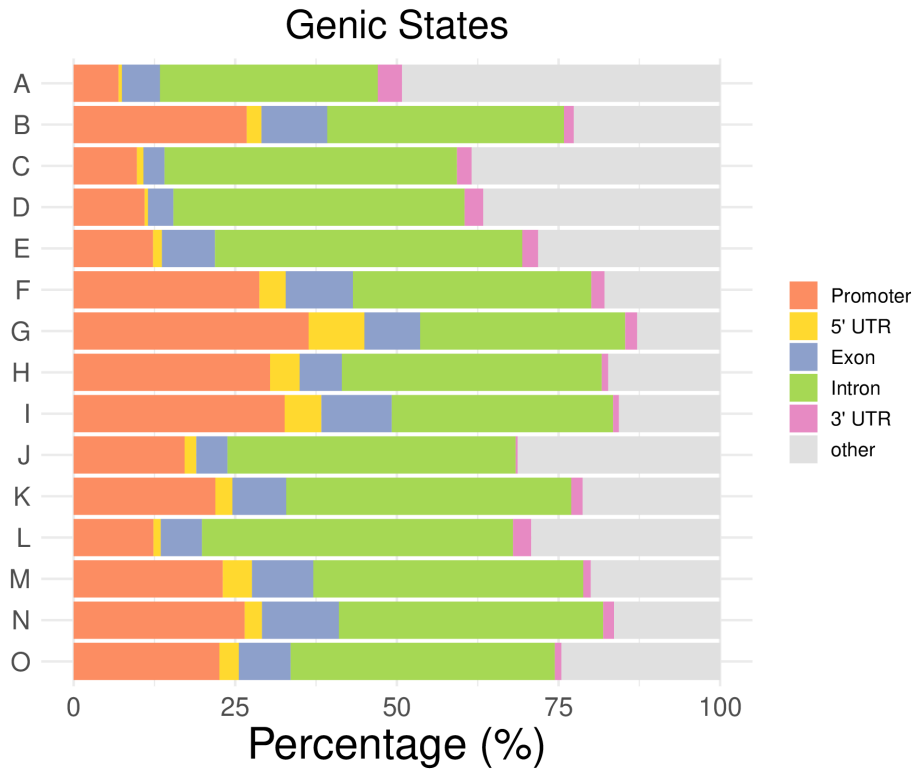


Figure 4. Genic annotation breakdown of 0hr region clusters. Percentages of clustered regions (A–O) overlapping promoters, 5' UTR, exons, introns, 3' UTR, or intergenic sequence are shown. Intronic annotations dominate across clusters, promoters are generally the next most abundant, UTRs are minimal, and the intergenic share varies with larger fractions in some clusters (e.g., A).

6 hr post-stimulation: Activation modules, enriched TF networks, and regulatory element composition

Biclustering of the 6 hr profiles resolved coherent activation modules dominated by a center-high, flanks-low architecture (Figure 5). In the majority of modules, the center column was strongly positive while the flanking columns were weak or near baseline, with several low-signal modules confined to the top of the matrix. The amplitude of the center signal varied across modules and was most pronounced in the lower half of the heatmap; no flanks-high, center-low inversions were evident. Together, these patterns indicate a strengthened, center-weighted activation architecture at 6 hr, with heterogeneous module-specific gain layered onto a common core.

TF motif analysis across clusters A–O showed a broadly elevated enrichment landscape that peaked in mid-to-late clusters, with a coordinated increase most apparent around H–K (Figure 6). Motifs for stimulus- and enhancer-associated regulators were prominent, including AP-1 (JUN/JUND), EP300, MAX/USF1/USF2, GABPA/ETS1, and RNA polymerase II (POLR2A), whereas early clusters (A–C) exhibited relative depletion for NANOG, MXI1, ZMIZ1, and CEBPB. The rise in AP-1 and EP300 coincides with

classical PMA-responsive chromatin programs and enhancer coactivator engagement observed in whole-blood stimulation models (Cayford et al., 2025). Enrichment for ETS-family and E-box factors (ETS1/GABPA; MAX/USF) further supports an activation network that integrates stimulus-responsive and broadly expressed transcriptional modules; the near-uniform PAX5 motif signal likely reflects underlying sequence composition rather than lineage restriction. Collectively, the mid-to-late clusters consolidate a TF network consistent with enhancer-driven, transcriptionally engaged states by 6 hr, relative to the weaker, early-cluster baseline.

Consistent with this network logic, cCRE annotations revealed enhancer predominance across all clusters (Figure 7). Distal enhancer-like signatures (dELS) comprised the largest fraction in every group, with proximal enhancer-like (pELS) and promoter-like (PLS) elements contributing substantially but secondarily. DNase-only and TF-associated (CA–TF) categories remained minor, and CA–H3K4me3 and CA–CTCF classes were minimal throughout. The modest variation between clusters, together with strong EP300/AP-1 motif enrichment, is congruent with an enhancer-centric activation program rather than a shift toward architectural (CTCF) or purely promoter-restricted control (Cayford et al., 2025).

Genic context mirrored this composition: 6 hr regions were predominantly intronic in all groups, with promoters the next most represented category and exons and UTRs comparatively underrepresented (Figure 8). Intronic fractions were highest in several mid-to-late clusters (e.g., B, H, L), while promoter contributions were notably elevated in D, E, H, L, and O; exonic content was comparatively higher in F, I, N, and O. These distributions indicate that the 6 hr program primarily engages intragenic regulatory elements—likely intronic enhancers—with selective promoter-proximal participation in subsets of clusters that also exhibit strong center-column activation and TF motif gain. Relative to the early-cluster baseline within the 6 hr stratification, the mid-to-late groups pair increased TF motif enrichment with subtly greater promoter involvement in specific modules, while maintaining a consistent, enhancer-dominated cCRE landscape.

In sum, by 6 hr post-stimulation the regulatory architecture is characterized by robust, center-high activation modules underpinned by an enhancer-centric cistrome and a TF network enriched for AP-1, EP300, ETS- and E-box-binding factors, mapping largely to

intronic elements with targeted promoter engagement in select clusters (Figures 5–8) (Cayford et al., 2025).

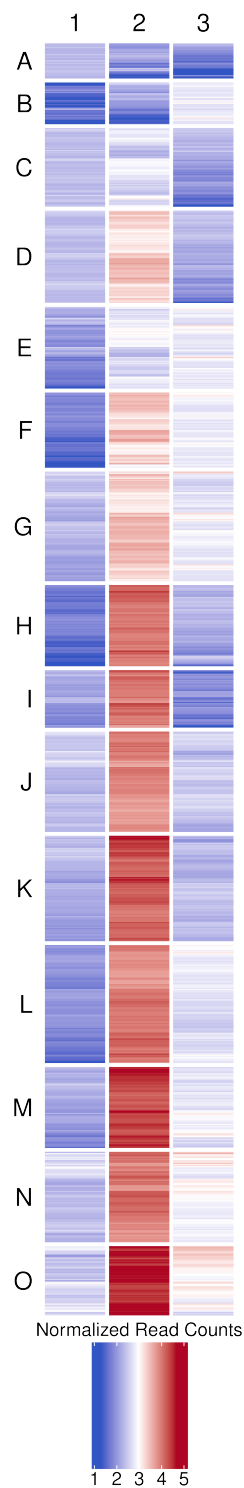


Figure 5. Biclustering highlights central feature activation at 6 hr. Genomic regions (rows) and three feature pairs (columns) are co-clustered by normalized signal at 6 hr. Most biclusters show elevated activity in the middle feature with lower levels in the flanking features, while a minority display uniformly low signals across all features.

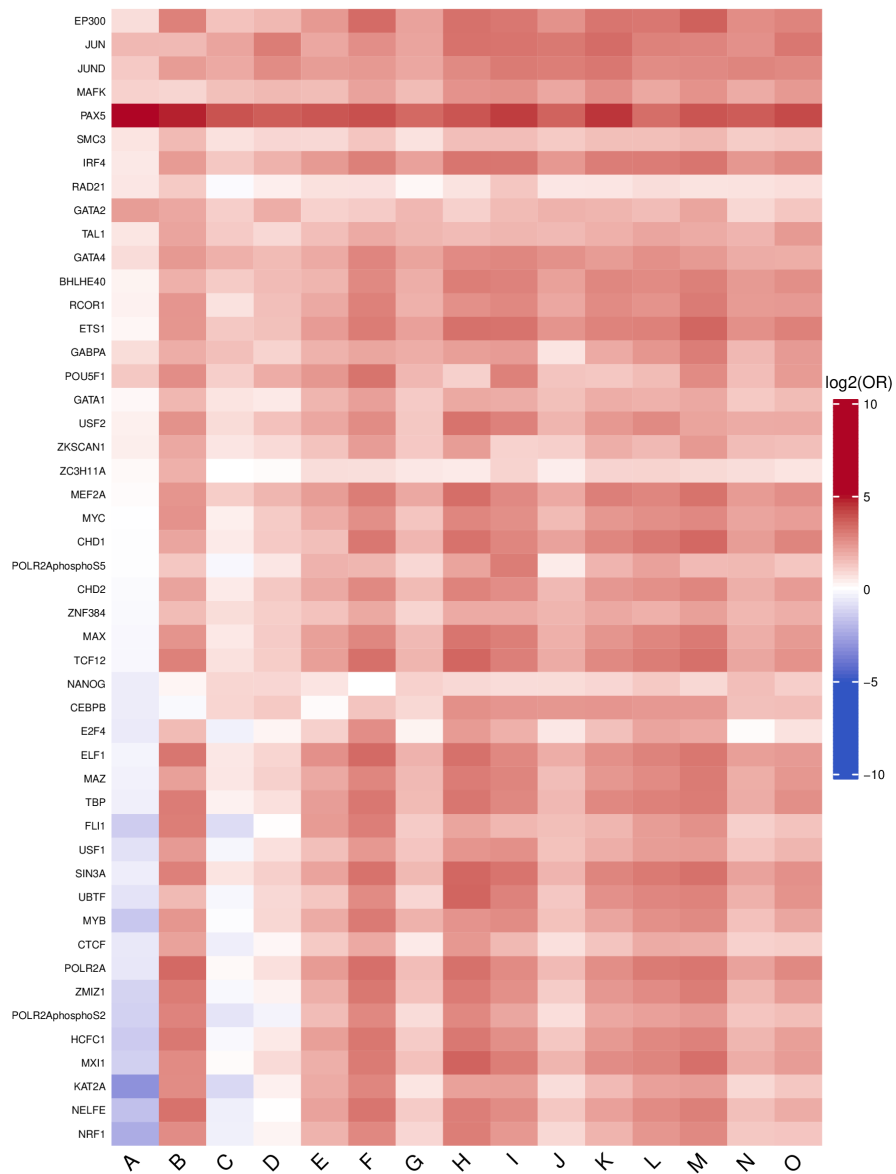


Figure 6. TF motif enrichment across 6hr region clusters. Motif overrepresentation is summarized for clusters A–O. Enrichment is widespread, with PAX5 highest across clusters and broad elevation for EP300, JUN/JUND, MAX, and POLR2A-associated motifs, while lower or negative signals for NANOG, MXI1, ZMIZ1, and CEBPB are concentrated in early clusters.

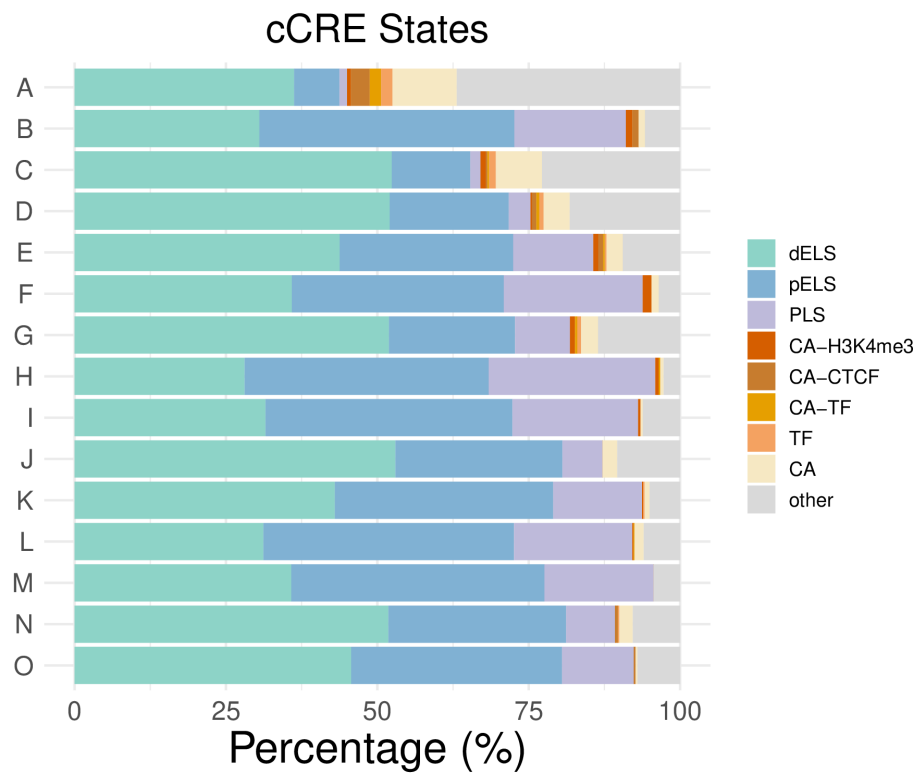


Figure 7. cCRE category composition across 6hr clusters. For each cluster (A–O), regions identified at 6hr are annotated with ENCODE cCRE categories to show their compositional percentages. Most clusters are dominated by dELS and pELS with a substantial PLS component, while CA- and TF-related states and the other category contribute smaller fractions.

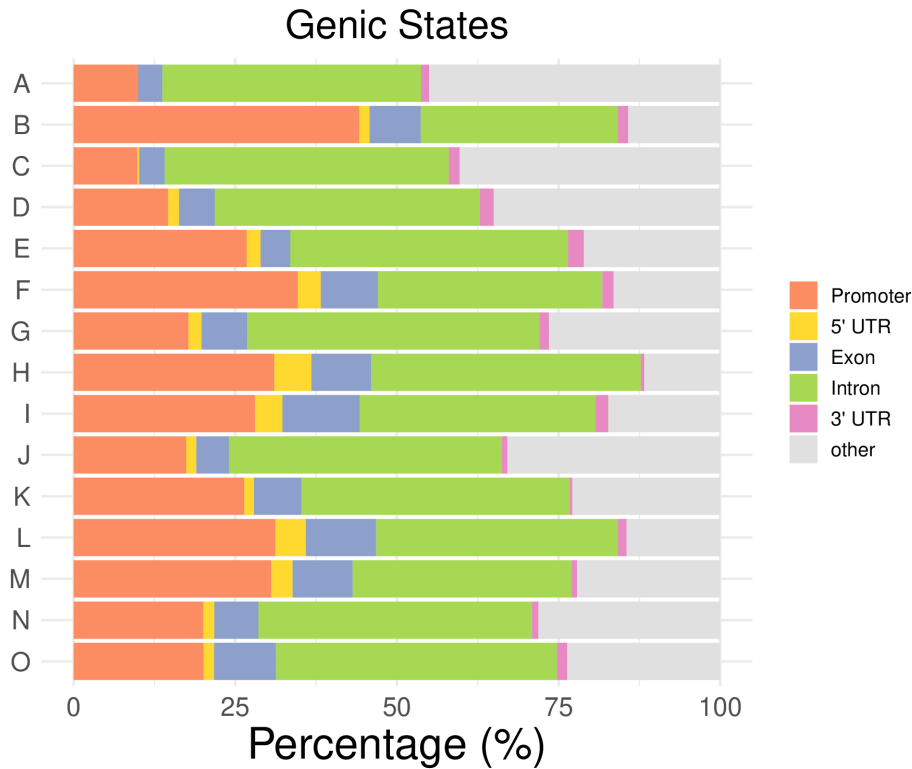


Figure 8. Genic annotation composition of 6hr regions across groups. *Composition of 6hr regions among promoter, exon, intron, UTR, and other categories for groups A–O. Introns dominate across groups, promoters are second most prevalent, exons vary by group, and both UTRs remain minor with a consistent other fraction present.*

24 hr: Sustained/late modules, TF and cohesin involvement, and genomic localization

Biclustering of 24 hr features resolved a late-time architecture dominated by center-column activation across many modules (Figure 9). Most biclusters exhibited high center-column signal coupled to markedly lower left-column signal and lower-to-moderate right-column signal, consistent with sustained/late activation of the assayed feature pair at this time point. A small set of modules near the top showed the opposite tendency, with reduced center-column signal and comparatively elevated right-column signal, whereas successive modules toward the bottom maintained strong center-column activity.

TF and cofactor motif enrichment was broad rather than polarized across clusters A–O, with several regulators showing pervasive enrichment (Figure 10). JUND (AP-1) displayed the strongest and most uniform signal across clusters, aligning with late inflammatory and chromatin-remodeling programs in which AP-1 components are engaged (Lin et al., 2023). Cohesin subunits RAD21 and SMC3 were consistently enriched, as were POLR2A phospho-Ser5 and phospho-Ser2, indicating coordinated enhancer–promoter engagement with concurrent RNA polymerase II initiation and elongation across sustained modules. The cohesive architecture inferred from RAD21/SMC3 enrichment is mechanistically

consonant with cohesin-facilitated enhancer–promoter looping observed in other systems (Ye et al., 2026). By contrast, several factors showed subgroup-specific attenuation: NRF1, ZMIZ1, KAT2A, and NELFE displayed weaker or negative enrichment in subsets, most apparent toward clusters A–C and the midline groups; NANOG and POU5F1 were only sparsely detected. A mild dip in overall TF enrichment was visible around clusters H–J, with stronger signals reappearing toward L–O. The variable NRF1 signal is notable given its role in mitochondrial gene regulation and oxidative metabolism programs (Park et al., 2026).

Genomic annotations were stable across clusters and dominated by enhancer-like elements (Figure 11). Distal enhancer-like signatures (dELS) comprised the largest share in every group, followed by proximal enhancer-like (pELS), with promoter-like sequences (PLS) present as a smaller but consistent fraction; CA–CTCF and CA–TF categories remained thin throughout. This enhancer-weighted cCRE composition is consistent with the widespread cohesin and POLR2A phosphorylation enrichments observed. Genic context similarly showed a consistent intron-dominant landscape with secondary promoter representation (Figure 12), but with cluster-level differences: A and C were particularly intron-heavy with little promoter content, B, D–F, H, and O exhibited comparatively larger promoter fractions, and G–J showed relatively elevated exon assignments. UTR categories remained small across all groups, and the “other” category was reduced in G–J relative to A or O.

Together, these data indicate that at 24 hr the chromatin program is organized into sustained/late modules characterized by strong center-column activation, enhancer-rich regulatory architecture, and pervasive occupancy by AP-1, cohesin, and transcriptionally engaged RNA polymerase II. Subtle midline dips (H–J) and subgroup differences (e.g., A–

C vs. L–O; promoter- vs. exon-leaning groups) modulate this otherwise uniform late-time regulatory landscape.

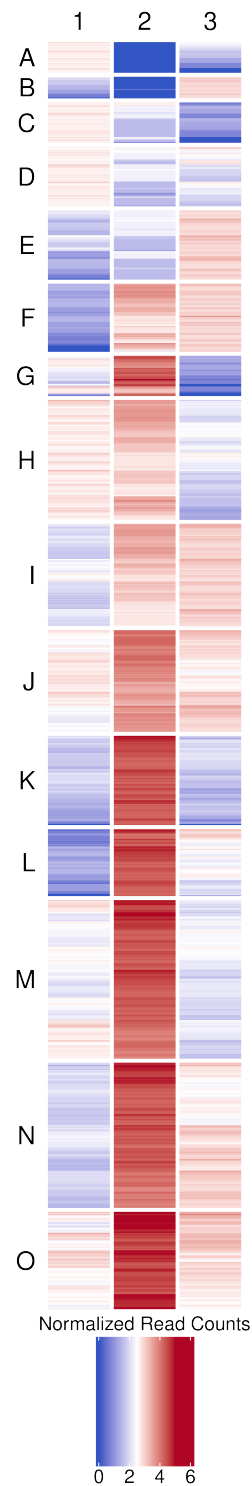


Figure 9. Biclustering reveals dominant middle-column activation at 24 hr. Heatmap of three feature pairs across clustered genomic regions from the 24 hr dataset. Most modules exhibit elevated signal in the middle feature with reduced signal in the left feature and variable levels in the right feature, highlighting coordinated modules at this time point.

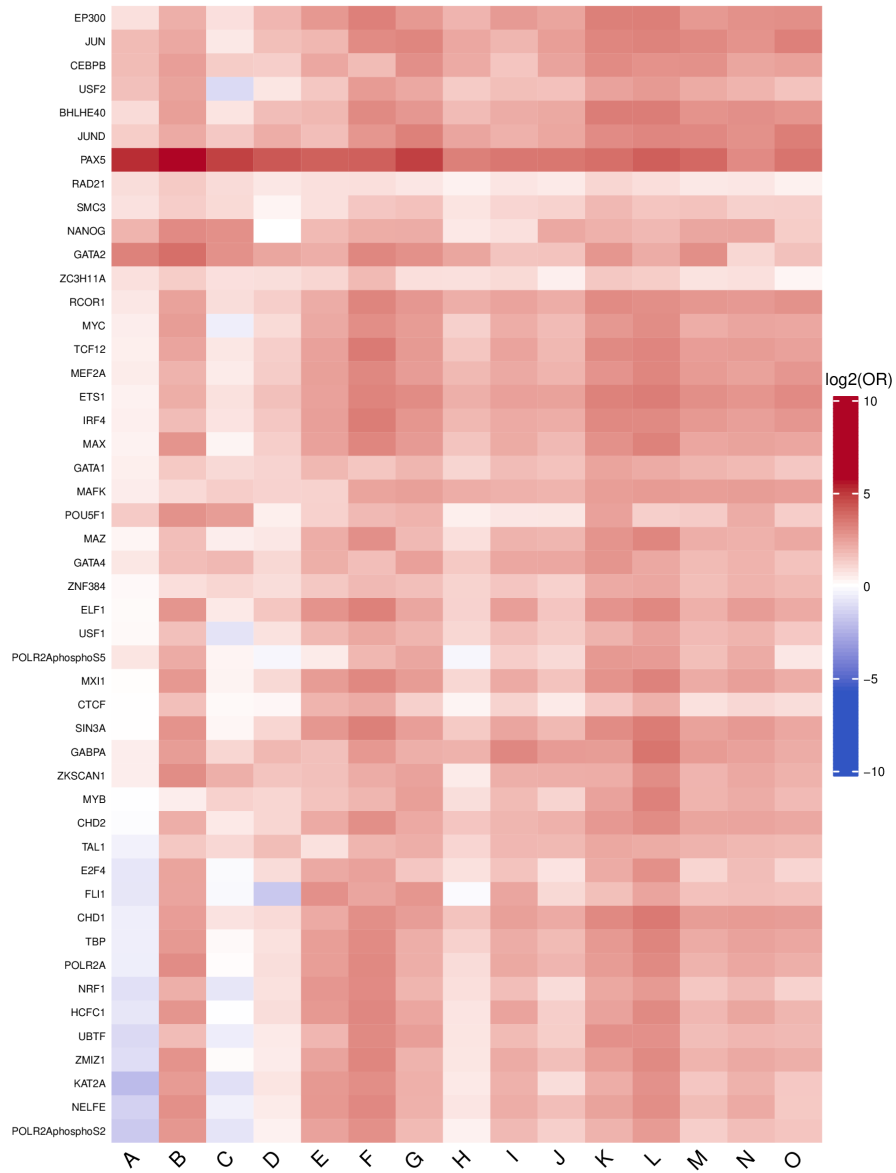


Figure 10. TF motif enrichment across 24hr region clusters. Motif overrepresentation is summarized for clusters A–O, highlighting broadly elevated signals for JUND, RAD21/SMC3, and POLR2A phosphorylation states, with patchier or reduced signals for NRF1, ZMIZ1, KAT2A, and NELFE in select clusters. These patterns nominate candidate regulators associated with the 24hr chromatin states.

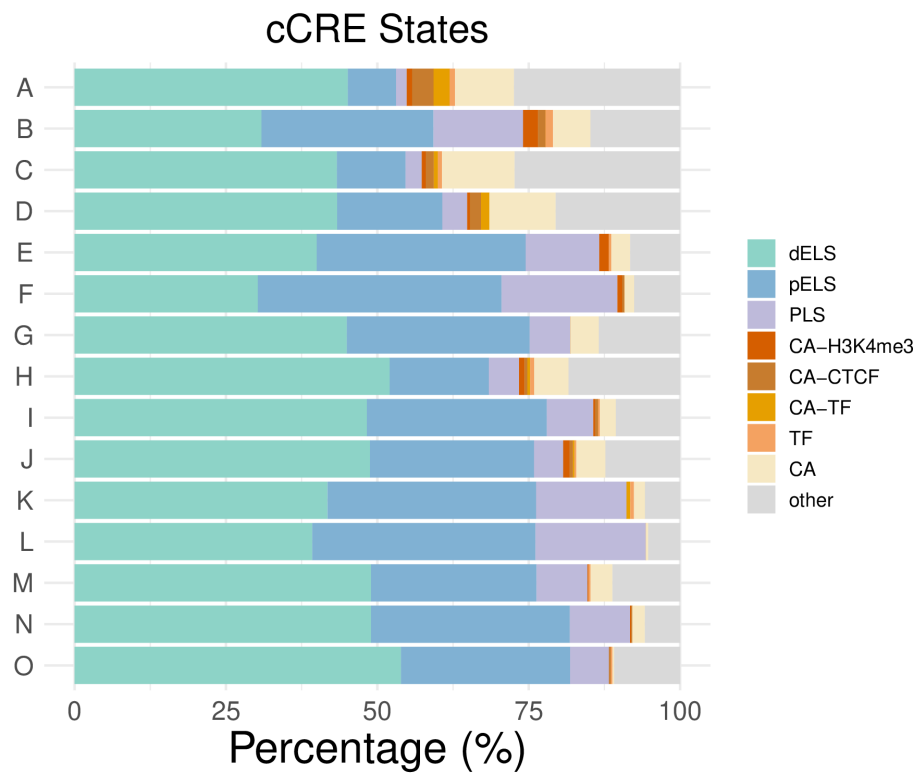


Figure 11. cCRE class composition of 24hr genomic clusters. Clusters A–O are summarized by their percentages across cCRE states (dELS, pELS, PLS, CA–H3K4me3, CA–CTCF, CA–TF, TF, CA, other). Enhancer-like states (dELS, pELS) predominate in every cluster, with PLS present at lower levels and the remaining categories contributing minor fractions with modest between-cluster variation.

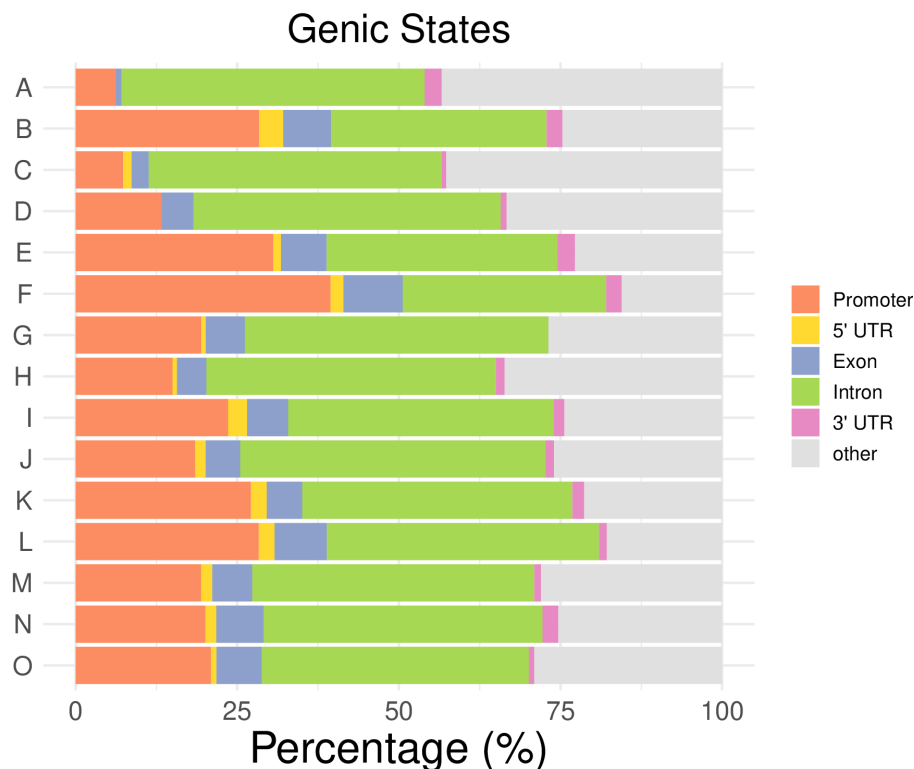


Figure 12. Genic distribution of 24hr regions across biclusters. Stacked proportions for biclusters A–O partition regions among promoter, 5' UTR, exon, intron, 3' UTR, and other categories. Most biclusters are intron-dominant with variable promoter contributions, while exon and UTR fractions remain comparatively small.

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